CHAPTER 07

# CI/ SI/ Instalments

Fundamentally, this chapter is the easiest one. It requires basic understanding of percentage and ratio-proportion along with the knack for simple calculations. If ever any problem is asked from this chapter, it is hardly a tricky one. That's why CAT is not a great fan of this chapter. However, this chapter holds relevance for most of the competitive exams and even for the Data Interpretation section. Nonetheless, Banking exams and SSC CGL exams and recruitment exams ask a significant number of problems from this chapter. In order to score more, we must rely on common sense and logic than that on formulae.

## 7.1 Important Definitions

**Simple interest** is nothing but the fix percentage of the principal (invested/borrowed amount of money). Some key words used in the concept of interest.

**Principal** ($P$)**:** It is the sum of money deposited/loaned etc. also known as **capital.**

**Interest :** It is the money paid by borrower, calculated on the basis of **Principal.**

**Time** ($T/n$) : This is the duration for which money is lent/borrowed.

**Rate of Interest** ($r/R$) : It is the rate at which the interest is charged on principal.

**Amount** ($A$) = Principal + Interest

**Simple Interest :** When the interest is calculated uniformly only on the principal for the given time period.

**Compound Interest :** In this case for every next period of time the interest is charged on the total previous amount (which is the sum of principal and interest charged on it so far.) *i.e.,* every time we calculate successive increase in the previous amount.

**Chapter Checklist**


- *Important Definitions*
- *Simple Interest*
- *Compound Interest (CI)*
- *Depreciation*
- *Calculating the Population*
- *Instalments*
- *Important Amounts, Interest Rates and Time Periods*
- ***CAT Test***


## 7.2 Simple Interest (SI)

If the Principal (P), Rate of Interest (r) and Time Period (t) are given, we will have

$$\text{Simple Interest (SI)} = \frac{P \times r \times t}{100}$$

$$\text{Amount (A)} = P + \frac{Prt}{100} = P\left(1 + \frac{rt}{100}\right)$$

**NOTE** Out of the five variables $A$, SI, $P$, $r$, $t$ we can find any one of these, if we have the requisite information about the four variables directly or indirectly individuality or jointly.

### *Conversion of Time Period and rate of Interest*

| Given ($r$%) | Given ($t$) | Required ($r$%) | Required ($t$) |
|---|---|---|---|
| $r$% annual | $t$ years | $\frac{r}{2}$(%) half-yearly | $2t$ |
| $r$% annual | $t$ years | $\frac{r}{4}$% quarterly | $4t$ |
| $r$% annual | $t$ years | $\frac{r}{12}$% monthly | $12t$ |

## 7.3 Compound Interest (CI)

1. CI $= A - P$
   [$A$ is the amount including interest and principal ($P$) both]
2. Amount $(A) = P\left(1+\frac{r}{100}\right)^t$
3. When the rate of interest is half-yearly $A = P\left(1+\frac{(r/2)}{100}\right)^{2t}$
4. When the rate of interest is quarterly $A = P\left(1+\frac{r/4}{100}\right)^{4t}$
5. Difference between CI and SI for two years $= P\left(\frac{r}{100}\right)^2$
6. Difference between CI and SI for three years
$$= P\left(\frac{r}{100}\right)^2\left(\frac{r}{100}+3\right)$$
7. Difference between CI and SI for $n^{\text{th}}$ year
$$= \frac{Pr}{100}\left[\left(1+\frac{r}{100}\right)^{n-1}-1\right]$$
8. For compound interest, if $r$ denotes the rate of interest, the change in amount over the previous year can be calculated or following.
$$\frac{\text{increase in amount in } n^{\text{th}} \text{ year}}{\text{increase in amount in } (n+1)^{\text{th}} \text{ year}} = \frac{100}{(100+r)}$$
Similarly,
$$\frac{\text{decrease in amount in } n^{\text{th}} \text{ year}}{\text{decrease in amount in } (n+1)^{\text{th}} \text{ year}} = \frac{100}{(100-r)}$$
where $r$ is rate of interest.

## 7.4 Depreciation

It is known that the prices of some articles depriciates in their values over a time period. When the value in terms of currency decreases, we say that the value of the article is depreciating $V_f = V_i\left(1-\frac{r}{100}\right)^t$.

$V_i$ = Initial value of the article

$V_f$ = Final (depreciated) value of article $r$, is the rate of interest by which the price of article decreases over the time period '$t$'.

## 7.5 Calculating the Population

It has been observed that the population of a particular locality/nation etc. increases or decreases successively over its previous value *i.e.,* it increases or decreases as compound interest for the money. Thus we use similar formulae for the calculation of population.

Total population $(P) = P_0\left(1+\frac{r}{100}\right)^n$, when population increases. Total population $(P) = P_0\left(1-\frac{r}{100}\right)^n$, when population decreases.

## 7.6 Instalments

When a borrower paid the sum in parts (*i.e.,* not in a single amount) then we say that he/she is paying in instalments. For example $A$ borrowed ₹ 100 from $B$ and he pays back it to $B$ in several parts *i.e.,* ₹ 20 in 5 times or ₹ 50 in 2 times etc. But the important point is that borrower has to also pay the interest for using the borrowed sum/or purchased article. In general the value of each instalment is kept constant even when the interest charged on each instalment vary for each instalment.

### *For Simple Interest*

Let A be the total amount to be paid, $r$ be the rate of interest, $x$ be the value of each instalment and $n$ be the number of instalments, we have

$$\text{A} = \left[x+\left(x+\frac{x\times r\times 1}{100}\right)+\left(x+\frac{x\times r\times 2}{100}\right)+\left(x+\frac{x\times r\times 3}{100}\right)+\ldots+\left(x+\frac{x\times r\times n}{100}\right)\right]$$

Let P be the principal amount, we have

$$\text{A} = \text{P}+\frac{\text{P}\times\text{n}\times\text{r}}{100}$$

Therefore, we have

$$\text{P}+\frac{\text{P}\times\text{n}\times\text{r}}{100} = \left[x+\left(x+\frac{x\times r\times 1}{100}\right)+\left(x+\frac{x\times r\times 2}{100}\right)+\left(x+\frac{x\times r\times 3}{100}\right)+\ldots+\left(x+\frac{x\times r\times n}{100}\right)\right]$$

### *For Compound Interest*

Let P be the principal amount, $r$ be the rate of interest, $x$ be the value of each instalment and $n$ be the number of instalments, we have

$$P = \left[\frac{x}{\left(1+\frac{r}{100}\right)} + \frac{x}{\left(1+\frac{r}{100}\right)^2} + \frac{x}{\left(1+\frac{r}{100}\right)^3} + \ldots + \frac{x}{\left(1+\frac{r}{100}\right)^n}\right]$$

Let A be the total amount paid back in $n$ instalments, we have

$$A = P\left(1+\frac{r}{100}\right)^n$$

Therefore, we have

$$P\left(1+\frac{r}{100}\right)^n = \left[x + x\left(1+\frac{r}{100}\right) + x\left(1+\frac{r}{100}\right)^2 + \ldots + x\left(1+\frac{r}{100}\right)^n\right]$$

**Exp. 1)** Find the simple interest on ₹ 1000 at 12% per annum for 5 years

**Solution** $SI = \frac{Prt}{100} = \frac{1000\times 12\times 5}{100} = ₹\ 600$

{Total amount = $P$ + SI = 1000 + 600 = ₹ 1600}

**Exp. 2)** Find the simple interest on ₹ 800 at 7% per annum and on ₹ 700 at 16% p.a. and on ₹ 500 at 4% p.a. for 2 years

**Solution** $SI = \frac{P_1 r_1 t_1}{100} + \frac{P_2 r_2 t_2}{100} + \frac{P_3 r_3 t_3}{100}$

$= \frac{800\times 7\times 2}{100} + \frac{700\times 16\times 2}{100} + \frac{500\times 4\times 2}{100}$

$= 112 + 224 + 40 = ₹\ 376$

**Exp. 3)** A sum of money ($P$) doubles in 10 year In how many years it will be treble at the same rate of simple interest?

**Solution** $A = 2P$

$\therefore$ $SI = P$ $\quad (SI = 2P - P)$

$P = \frac{P\times r\times 10}{100} \Rightarrow r = 10\%$

So, the new amount = $3P$

But the new $SI = 2P = (3P - P)$; $2P = \frac{P\times 10\times t}{100}$ $\quad (r = 10\%)$

$t = 20$ years

**Exp. 4)** A sum of money in 3 years becomes 1344 and in 7 years it becomes ₹ 1536. What is the principal sum where simple rate of interest is to be charged ?

(a) 4000 (b) 1500
(c) 1200 (d) 2800

**Solution** It would be very time saving if we do it by unitary method.

1536 – 1344 = ₹ 192

₹ 192 is the interest for 4 years

So, ₹ 48 is the interest for 1 year

It means ₹ 144 is the interest for 3 years at actual principal.

$\therefore$ Principal = 1344 – 144 = ₹ 1200

**Exp. 5)** A certain sum of money amounts to ₹ 15900 at simple rate of interest at 6% p.a. in 1 year. What is the value of principal sum ?

(a) 12000 (b) 18000 (c) 15000 (d) 14000

**Solution** Best way is to go through option as $\frac{15000\times 6\times 1}{100} = 900$

So, the total amount = 15900, hence (c) is correct.

**Exp. 6)** A sum of money becomes 3 times in 12 years. In how many years it will become 5 times at the same rate of simple interest?

(a) 20 years (b) 16 years (c) 24 years (d) 30 years

**Solution** In this type of question consider only SI not the amount. So, (in first case) SI = 2$P$ for principal $P$

(in second case) SI = 4$P$ for principal $P$.

Now, since SI is directly proportional to the time at fixed rate of interest.

Therefore, to make SI two times it will require 12 years

So, to make SI four times it will require 24 years

Thus to make an amount 5 times of the principal requires 24 years at the given rate of interest as per question.

**Exp. 7)** What will be the amount when ₹ 10000 is deposited in a bank at 10% per annum compounded annually for 3 years?

**Solution** $A = P\left(1+\frac{r}{100}\right)^n = 10000\left(1+\frac{10}{100}\right)^3$

$= 10000\times(1.1)^3 = ₹\ 13310$

**Exp. 8)** A sum of amount at $r$% compound interest doubles in 3 years. In 9 years it will be $k$ times of the original principal. What is the value of $k$?

(a) 10 (b) 9 (c) 6 (d) 8

**Solution** $A = P\left(1+\frac{r}{100}\right)^n$

1st case $2P = P\left(1+\frac{r}{100}\right)^3 \Rightarrow 2 = \left(1+\frac{r}{100}\right)^3$

2nd case $(2)^3 = \left[\left(1+\frac{r}{100}\right)^3\right]^3 = \left(1+\frac{r}{100}\right)^9 \Rightarrow k = (2)^3 = 8$ times.

**Exp. 9)** ₹ 20000 is being compounded at 20% p.a. If the rate of interest is charged half yearly. What will be the amount after 2 years ?

(a) 28292 (b) 27292
(c) 29282 (d) 22358

**Solution** $20000\left(1+\frac{20/2}{100}\right)^{2\times 2} = 20000\left(1+\frac{10}{100}\right)^4$

$= 20000\,(1.1)^4 = 20000\times 1.4641$

$= ₹\ 29282$

**Exp. 10)** If the rate of interest is 10% p.a. and ₹ 12000 lent at the compound interest, half yearly. What is the equivalent simple rate of interest for first year?

**Solution** $A = P\left(1+\frac{r}{100}\right)^n = 12000\,(1.05)^2$

$= 12000\times 1.1025 = ₹\ 13230$

Again $SI = 13230 - 12000 = 1230$

$\therefore \quad 1230 = \frac{12000\times r\times 2}{2\times 100}$

$\Rightarrow \quad r = 10.25\%$

**Exp. 11)** ₹ 1000 is being charged at 50% per annum. What is the interest for $3^{rd}$ year at compound interest ?

(a) 1122 (b) 1025
(c) 1125 (d) 625

**Solution** Total amount for two years $= 1000\,(1.5)^2 = 2250$

Now, interest for third year $= \frac{2250\times 50\times 1}{100} = 1125$

Hence, (c) is correct.

**Exp. 12)** CI and SI for a certain sum at certain rate of interest are ₹ 220 and ₹ 200. What is the principal for these interests?

(a) ₹ 2200 (b) ₹ 200
(c) ₹ 500 (d) ₹ 2000

**Solution** $220 - 200 = 20$

₹ 20 is the interest on the interest of the first year, which is ₹ $100 = \left(\frac{200}{2}\right)$

Thus, the rate of interest = 20% $\left(20 = \frac{100\times r\times 1}{100}\right)$

So, the actual principal = $P = 500$

$\left(\text{since } 100 = \frac{P\times 20\times 1}{100} \Rightarrow P = 500\right)$

Hence, (c) is correct.

**Exp. 13)** The population of UP increases very rapidly @ 12% p.a. and 3 years ago its population was 1 million. What is present population?

**Solution** $A = 1000000\,(1.12)^3 = 1404928$

**Exp. 14)** Jadeja purchased a maruti car 3 years ago for ₹ 2 lakh. Its value depreciated each year @ 25% p.a. What is the present value of the car ?

**Solution** $200000\left(1-\frac{25}{100}\right)^3 = 200000\times\left(\frac{3}{4}\right)^3 = 84,375$

**Exp. 15)** The difference between CI and SI for 3 years @ 20% p.a. is ₹ 152. What is the principal lent in each case ?

(a) 1200 (b) 1155
(c) 1187.5 (d) none of these

**Solution** Difference between CI and SI for 3 years = ₹ 152

$$p\left(\frac{r}{100}\right)^2\left(\frac{r}{100}+3\right) = 152$$

$$P\left(\frac{1}{25}\right)\left(\frac{16}{5}\right) = 152 \Rightarrow P = \frac{152\times 25\times 5}{16}$$

$$P = 9.5\times 25\times 5 = 1187.5$$

**Exp. 16)** A sum is being lent at 20% p.a. compound interest. What is the ratio of increase in amount of $4^{th}$ year to $5^{th}$ year?

(a) 4 : 5 (b) 5 : 4
(c) 5 : 6 (d) can't be determined

**Solution** $\frac{P\left(1+\frac{r}{100}\right)^4}{P\left(1+\frac{r}{100}\right)^5} = \frac{1}{\left(1+\frac{r}{100}\right)} = \frac{100}{100+r} = \frac{100}{120} = \frac{5}{6}$

**Exp. 17)** ₹ 12000 amounts to ₹ 20736 in 3 years at $r$% p.a. of compound interest. What is the value of $r$?

(a) 10% (b) 25% (c) 12% (d) 20%

**Solution** $A = P\left(1+\frac{r}{100}\right)^3 \Rightarrow 20736 = 12000\left(1+\frac{r}{100}\right)^3$

$\Rightarrow \frac{20736}{12000} = \left(1+\frac{r}{100}\right)^3 \Rightarrow \frac{1728}{1000} = \left(1+\frac{r}{100}\right)^3$

$\Rightarrow \left(\frac{12}{10}\right)^3 = \left(1+\frac{r}{100}\right)^3 \Rightarrow \left(1+\frac{2}{10}\right)^3 = \left(1+\frac{r}{100}\right)^3 \Rightarrow r = 20\%$

**Alternatively**

The best way for this problem is to go through options.

$12000\times 1.2 \Rightarrow 14400\times 1.2 \Rightarrow 17280\times 1.2 \Rightarrow 20736$

**Exp. 18)** A certain sum amounts to ₹ 14641 in 4 years @ 10% p.a. compounded annualy. What is the value of principal?

(a) ₹ 6000 (b) ₹ 12000
(c) ₹ 10000 (d) data insufficient

**Solution** $14641 = P\left(1+\frac{10}{100}\right)^4 \Rightarrow 14641 = P\left(\frac{11}{10}\right)^4$

$$P = 14641\times\left(\frac{10}{11}\right)^4 = 10000$$

# CAT-Test

**Questions Helping you bell the CAT**

## LEVEL 01 > BASIC LEVEL EXERCISE

**1** A man borrows ₹ 4000 and pays back after 5 years at 15% simple interest. The amount paid by the man is :
(a) ₹ 1800 (b) ₹ 4800
(c) ₹ 7500 (d) ₹ 7000

**2** What is the time period for which ₹ 8000 amounts to ₹ 12000 at 20% p.a. of simple interest?
(a) 4 years (b) 2.5 years
(c) 3.25 years (d) 6 years

**3** What is the rate of simple interest at which ₹ 14,000 gives interest of ₹ 1960 in two years?
(a) 4% (b) 5%
(c) 7% (d) 10%

**4** What is the sum of amount which gives ₹ 6300 as interest @ 7% per annum of simple interest in $7\frac{1}{2}$ years?
(a) 36000 (b) 24000
(c) 63000 (d) 12000

**5** If the rate of simple interest is 12% per annum, the amount that would fetch interest of ₹ 6000 per annum is :
(a) ₹ 7200 (b) ₹ 48000
(c) ₹ 50000 (d) ₹ 72000

**6** A sum was put at simple interest at a certain rate for 2 years Had it been put at 4% higher rate, it would have fetched ₹ 112 more. The sum is :
(a) 1120 (b) 1400
(c) 1200 (d) 8000

**7** In what time will a sum of money double itself @ 20% per annum (p.a.) simple interest?
(a) 10 years (b) 5 years
(c) 2 years (d) 4 years

**8** At $r$% per annum a sum doubles after 20 years. The rate of interest per annum is :
(a) 4% (b) 5%
(c) 8% (d) 10%

**9** A sum of money trebles (i.e., 3 times) in 15 years at $r$% of simple interest per annum. What is the value of $r$?
(a) 12% (b) $\frac{40}{3}$%
(c) $\frac{50}{3}$% (d) can't be determined

**10** A sum of money doubles itself in 12 years. In how many years would it treble itself?
(a) 36 years (b) 18 years
(c) 24 years (d) 15 years

**11** Out of a sum of ₹ 625, a part was lent at 5% SI and the other at 10% SI. If the interest on the first part after 2 years is equal to the interest on the second part after 4 years, then the second sum (in ₹) is :
(a) 250 (b) 300
(c) 125 (d) 275

**12** A sum of ₹ 2500 is lent out in two parts; one at 12% p.a. and another at 12.5% p.a. for one year. If the total annual income is ₹ 306, the money lent at 12% is :
(a) 1000 (b) 1200 (c) 1500 (d) 1300

**13** Akul lent ₹ 6000 to Bakul for 2 years and ₹ 1500 to camlin for 4 years and received altogether from both ₹ 900 as simple interest. The rate of interest is :
(a) 4% (b) 8% (c) 10% (d) 5%

**14** A person takes a loan of ₹ 200 at 5% simple interest. He returns ₹ 100 at the end of one year. In order to clear his dues at the end of 2 years, he would pay :
(a) 125.50 (b) 110
(c) 115.50 (d) none of these

**15** A lends a sum of money for 10 years at 5% SI. *B* ends double the amount for 5 years at the same rate of interest. Which of the following statements is true in this regard?
(a) *A* will get twice the amount of interest that *B* would get
(b) *B* will get twice the amount of interest that *A* would get
(c) *A* will get the same amount of interest that *B* would get
(d) *B* will get four times the amount of interest that *A* would get

**16** Consider the following statements.
If a money is loaned at simple interest then the :
(i) money gets doubled in 5 years if the rate of interest is $16\frac{2}{3}$%.
(ii) money gets doubled in 5 years if the rate of interest is 20%.
(iii) money becomes four times in 10 years if it gets doubled in 5 years of these statements :

**15** $A$ and $B$ run a joint venture in which the profit earned by $A$ and $B$ are in the ratio 28 : 15. A invest his share at the start of the year and $B$ joins in after 9 months of the same year. What is the ratio of their initial investment, respectively?
(a) 7 : 15 (b) 8 : 13
(c) 5 : 17 (d) 15 : 7

**16** In the previous problem if $A$ gets a profit of ₹ 4200 the amount invested by $B$ is :
(a) 2250 (b) 2600
(c) 1350 (d) can't be determined

**17** Arvind and Govind each invested ₹ 15000 for 3 years at the same rate of interest but Arvind's investment is compounded annually while Govind's investment is charged on simple interest. What amount did Arvind receive more than Govind?
(a) ₹ 680 (b) ₹ 3450
(c) data insufficient (d) none of these

**18** Shyam Lal takes a loan of ₹ 10500 at 10% p.a. compounded annually which is to be repaid in two equal annual instalments. One at the end of one year and the other at the end of the second year. The value of each instalments is :
(a) 5987 (b) 6050
(c) 6352.5 (d) 5678.5

**19** Hari Lal and Hari Prasad have equal amounts. Hari Lal invested all his amount at 10% compounded annually for 2 years and Hari Prasad invested 1/4 at 10% compound interest (annually) and rest at $r$% per annum at simple interest for the same 2 years period. The amount received by both at the end of 2 year is same. What is the value of $r$?
(a) 14% (b) 12.5%
(c) 10.5% (d) 11%

**20** The annual sales of a company in the year 2000 was ₹ 1000 and in the year 2005 was ₹ 2490. Find the compounded annual growth rate (CAGR) of sales in the given period of the same company :
(a) 14.289% (b) 10%
(c) 15% (d) 20%

**21** HDFC lends 1 million to HUDCO at 10% simple interest p.a. for 2 years and HUDCO lends the same amount to SAHARA STATES HOUSING corporation at 10% p.a. of compound interest for 2 years What is the earning of HUDCO in this way?
(a) ₹ 133100 (b) ₹ 33100
(c) ₹ 131000 (d) no profit no loss

**22** ICICI lent ₹ 1 lakh to captain Ram Singh @ 6% per annum of simple interest for 10 years period. Meanwhile ICICI offered a discount in rate of interest for armed forces. Thus the rate of interest ICICI decreased to 4%. In this way Ram Singh had to pay total amount 1.48 lakh.
After how many year Ram Singh got the discount in rate of interest?
(a) 3 years (b) 4 years
(c) 6 years (d) 5 years

**23** Sanjay purchased a hotel worth ₹ 10 lakh and barkha purchased a car worth ₹ 16 lakh. The value of hotel every year increases by 20% of the previous value and the value of car every year depreciates by 25%. What is the difference between the price of hotel and car after 3 years?
(a) ₹ 925000 (b) ₹ 1053000
(c) remains constant (d) can't be determined

# Answers

## ✓ Level 01 *Basic Level Exercise*

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1. *(d)* | 2. *(b)* | 3. *(c)* | 4. *(d)* | 5. *(c)* | 6. *(b)* | 7. *(b)* | 8. *(b)* | 9. *(b)* | 10. *(c)* |
| 11. *(c)* | 12. *(d)* | 13. *(d)* | 14. *(c)* | 15. *(c)* | 16. *(c)* | 17. *(b)* | 18. *(b)* | 19. *(c)* | 20. *(b)* |
| 21. *(a)* | 22. *(a)* | 23. *(c)* | 24. *(c)* | 25. *(b)* | 26. *(c)* | 27. *(b)* | 28. *(a)* | 29. *(b)* | 30. *(b)* |
| 31. *(b)* | 32. *(c)* | 33. *(b)* | 34. *(c)* | 35. *(d)* | 36. *(c)* | 37. *(d)* | 38. *(c)* | 39. *(c)* | 40. *(b)* |
| 41. *(a)* | 42. *(d)* | 43. *(a)* | 44. *(b)* | 45. *(c)* | 46. *(b)* | 47. *(d)* | 48. *(b)* | 49. *(b)* | 50. *(b)* |
| 51. *(b)* | 52. *(b)* | 53. *(b)* | 54. *(c)* | 55. *(c)* | 56. *(d)* | | | | |

## ✓ Level 02 *Higher Level Exercise*

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1. *(b)* | 2. *(a)* | 3. *(d)* | 4. *(b)* | 5. *(d)* | 6. *(c)* | 7. *(c)* | 8. *(d)* | 9. *(c)* | 10. *(d)* |
| 11. *(a)* | 12. *(b)* | 13. *(d)* | 14. *(b)* | 15. *(a)* | 16. *(d)* | 17. *(c)* | 18. *(b)* | 19. *(c)* | 20. *(a)* |
| 21. *(c)* | 22. *(b)* | 23. *(b)* | | | | | | | |

# Hints & Solutions

## ☑ Level 01 *Basic Level Exercise*

**1** $4000 + (4000 \times 5 \times 0.15) = ₹\ 7000$

**2** $$(12000 - 8000) = \frac{8000 \times t \times 20}{100}$$

$$\Rightarrow \quad t = \frac{5}{2} \text{ years}$$

**3** $$1960 = \frac{14000 \times r \times 2}{100} \quad \Rightarrow \quad r = 7\%$$

**4** $$6300 = \frac{P \times 7 \times 15}{100 \times 2}$$

$$\Rightarrow \quad P = ₹\ 12000$$

**5** $$6000 = \frac{P \times 12 \times 1}{100}$$

$$\Rightarrow \quad P = 50000$$

**6** In my opinion this question should be solved by unitary method instead of making complex solution

| Years | Rate of Interest | Interest |
|---|---|---|
| 2 | 4% | 112 |
| 2 | 1% | 28 |
| 1 | 1% | (14) |

It means the principal sum is ₹ 1400

**Alternatively**

$$\frac{P}{100} \times 2[(r + 4) - r] = 112$$

$$\Rightarrow \quad P = 1400$$

**7** SI = $2P - P = P$ (Interest = Amount – Principal)

$$P = \frac{P \times 20 \times t}{100} \quad \Rightarrow \quad t = 5 \text{ years}$$

**8** $$P = \frac{P \times r \times 20}{100} \quad \text{(Interest = Amount – Principal)}$$

$$\Rightarrow \quad r = 5\% \text{ p.a.}$$

**9** $$2P = \frac{P \times 15 \times r}{100} \qquad (2P = 3P - P)$$

$$r = \frac{40}{3}\% \text{ p.a.}$$

**10** $$P = \frac{P \times r \times 12}{100} \quad \Rightarrow \quad r = \frac{100}{12}\% \text{ p.a.}$$

Now, $$2P = \frac{P \times 100 \times t}{12 \times 100} = 24 \text{ years}$$

**Shortcut (for SI)** Since SI in second case is double the SI in first case, so the time period will also be double since SI is directly proportional to the time period, provided that rate of interest be same and principal as well.

**11** $$\frac{P_1 \times 5 \times 2}{100} = \frac{P_2 \times 10 \times 4}{100}$$

$$\Rightarrow \quad P_1 : P_2 = 4 : 1$$

Therefore, second principal is ₹ $125 \left(= 625 \times \frac{1}{5}\right)$

**Alternatively** Go through options.

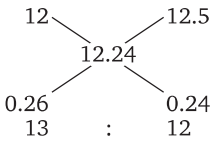
**12** Best way is to go through option

So, $1300 \times 0.12 + 1200 \times 0.125 = 306$

Hence, (d) is the correct option.

**Alternatively** Solve through alligation

Therefore the ratio of first principal to the second principal is 13 : 12.




$$306 = \frac{2500 \times r \times 1}{100}$$

$$r = \frac{306}{25} = 12.24$$

**13** $$\left(\frac{6000 \times 2 + 1500 \times 4}{100}\right) r = 900$$

$$\Rightarrow \quad r = 5\%$$

**14** Amount to be paid in first year $= \frac{200 \times 5 \times 1}{100} + 200 = 210$

Amount left as a principal for the second year

$$= 210 - 100 = 110$$

∴ Amount paid in second year

$$= 110 + \frac{110 \times 5 \times 1}{100} = 115.5$$

**15**

| *A* | *B* |
|---|---|
| $P_A = P$ | $P_B = 2P$ |
| $r_A = 5\%$ | $r_B = 5\%$ |
| $t = 10$ years | $t = 5$ years |
| Interest of $A = \frac{P \times 5 \times 10}{100}$ | Interest of $B = \frac{2P \times 5 \times 5}{100}$ |
| $= \frac{P}{2}$ | $= \frac{P}{2}$ |

Hence (c) is correct

**16** **Statement (I)** $\frac{P \times 5 \times 50}{100 \times 3} \neq P$, hence wrong

**Statement (II)** $\text{SI} = \frac{P \times 5 \times 20}{100} = P$, hence correct

**Statement (III)** $P = \frac{P \times 5 \times r}{100} \quad \Rightarrow \quad r = 20\%$

Again $\text{SI} = \frac{P \times 10 \times 20}{100} = 2P$, hence wrong